



AI FITTRACK

Personal Fitness Management System

AI-Augmented Backend Development Project Documentation

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BACHELOR OF COMPUTER APPLICATIONS

PROJECT TEAM		
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TECHNOLOGY OVERVIEW	
Project Type	Full-Stack Fitness Management System
Project Folder	AI-FITTRACK-API
Frontend	HTML + CSS + JavaScript
Backend	Node.js + Express.js
Database	MongoDB Atlas
AI Integration	AI-powered fitness recommendation module

Project documentation containing implementation details, testing evidence and working-project screenshots.

1. Project Description

AI FitTrack is a personal fitness management system developed as a full-stack web application. It combines user authentication, progress tracking, workout management, food and nutrition records, dashboard statistics and an AI-powered recommendation module. The browser-based frontend communicates with a Node.js and Express.js REST backend, while MongoDB Atlas provides persistent storage. The system is designed to centralize fitness information and make routine tracking easier through organized records, automatic BMI calculation and recommendation support.

The project uses HTML, CSS and JavaScript on the frontend; Node.js, Express.js and Mongoose on the backend; MongoDB Atlas for data storage; JWT-based authentication and bcryptjs for login security; and Thunder Client for API testing. The actual project repository is organized into client, config, controllers, middleware, models, routes and services modules. This reflects the implementation documented in the supplied AI FitTrack project report.

2. Scenario

Problem: Maintaining fitness information manually can make it difficult to remember workout history, monitor body measurements and review food information. Users may also find it difficult to understand overall progress when information is stored in separate places.

AI / Intelligent Support: AI FitTrack centralizes fitness records and provides an AI-powered recommendation section. The application also automates BMI calculation and dashboard summaries so that users can review important fitness information in one place.

Impact: The application is intended for personal fitness tracking. It helps users record progress, workouts and food details, review historical information and receive summarized dashboard information. It is a fitness management aid and does not replace professional medical diagnosis or treatment.

Important template alignment: The supplied template lists AI chatbot, content generation, question answering, semantic search, vector search, document analysis and RAG as examples. The current AI FitTrack report supports an AI recommendation module, but does not document those other capabilities as implemented. Therefore they are not falsely presented as completed features in this report.

3. System Requirements

Software Requirements

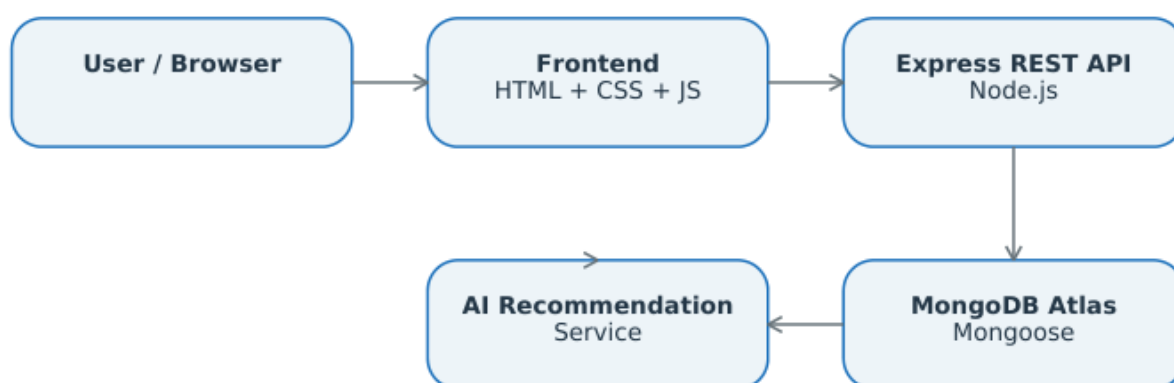
- Windows 10/11 or another supported operating system.
- Visual Studio Code.
- Node.js and npm.
- Express.js backend.
- MongoDB Atlas.
- MongoDB/Mongoose libraries.
- A modern web browser.
- Thunder Client for REST API testing.
- Git & GitHub for version control.

Hardware Requirements

- A computer capable of running VS Code and Node.js.
- At least 8 GB RAM is recommended.
- Sufficient storage for the project, dependencies and database tools.
- Internet connection for MongoDB Atlas and AI-related services.

4. Project Architecture

AI FitTrack follows a layered client-server architecture. The browser frontend sends HTTP requests to the Express backend. The backend validates requests, performs business operations and communicates with MongoDB Atlas through Mongoose. Authentication-protected requests carry a JWT token. The backend returns JSON responses which the frontend uses to update the interface.

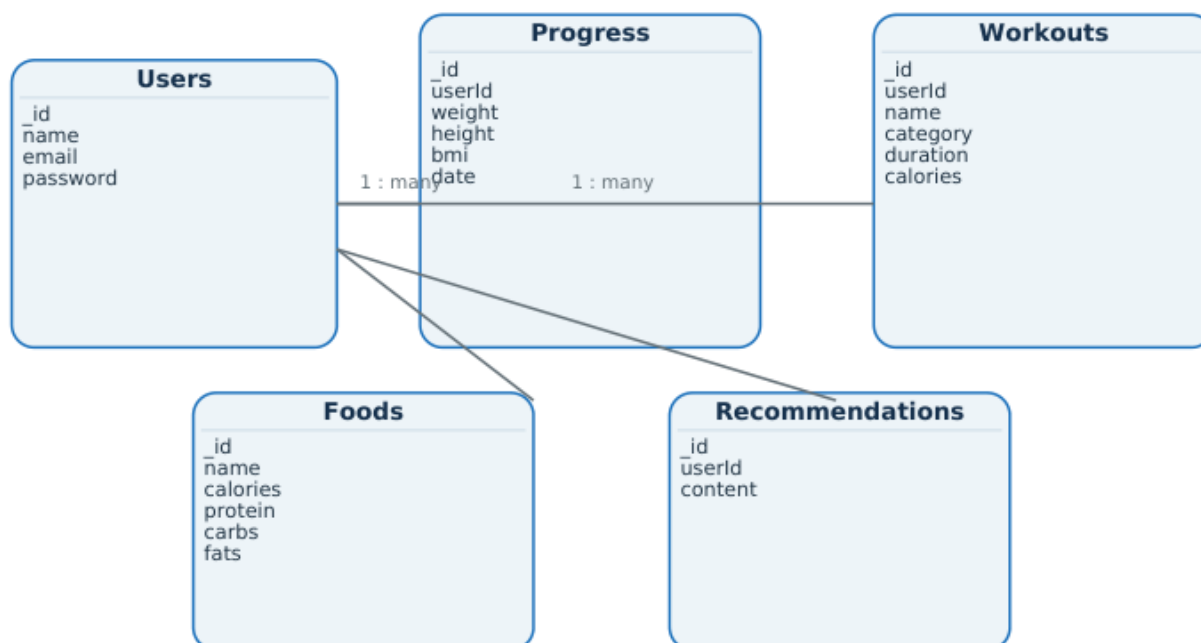


High-level flow

- User → Frontend interface
- Frontend → REST API request
- Express server → Route and controller
- Authentication middleware → Token validation
- Controller/service → Business operation
- Mongoose → MongoDB Atlas
- Backend → JSON response
- Frontend → Display updated result

5. ER Diagram and Data Design

The actual working database documented for AI FitTrack contains collections for users, workouts, progresses, foods and recommendations. A user can have multiple workout and progress records. Food and recommendation collections support the nutrition and recommendation portions of the application.



Collections

- users – authentication and user information.
- workouts – workout records.
- progresses – weight, height, BMI and date records.
- foods – nutrition and serving information.
- recommendations – recommendation information.

6. Features

Core Application Features

- User registration and login.
- JWT-based protected backend operations.
- Progress tracking with automatic BMI calculation.
- Progress history with edit and delete operations.
- Workout create, view, search, update and delete.
- Food create, view, search, update and delete.
- Dashboard statistics for workouts, calories and foods.
- BMI category and recommendation information.
- AI recommendation display.
- MongoDB Atlas persistent storage.

Template AI Feature Alignment

Template functionality	Current AI FitTrack status
MVC / modular architecture	Implemented through models, controllers, routes, middleware and services.
JWT Authentication	Implemented.
Semantic Search	Not documented as implemented in the current project.
MongoDB Atlas Search	Not documented as implemented.
MongoDB Atlas Vector Search	Not documented as implemented.
Prompt Engineering	Not documented as a separate implemented feature.
AI API Integration	AI recommendation service is present; exact provider/API details are not specified
Role-Based Access Control	Not documented as implemented; authentication/protected operations are implemented

7. Roles and Responsibilities

Current implementation

- Authenticated User – registers, logs in, manages progress, workouts and food information, views dashboard results and receives recommendation information.

The supplied template includes separate Admin and User roles with role-based access control. The current AI FitTrack project report does not document a separate Admin role or RBAC implementation, so those capabilities are not claimed as completed.

8. User Flow

- User Registration
- User Login
- JWT authentication
- Dashboard access
- Add or retrieve progress
- Automatic BMI calculation
- View progress history
- Add/search/update/delete workouts
- Add/search/update/delete food records
- View dashboard statistics
- View AI recommendation
- Logout / end session

9. MVC Pattern / Modular Backend Structure

The backend is organized in an MVC-style modular structure. The supplied project contains models, controllers, routes, middleware and services. This separation keeps authentication, database operations, business logic and API routing organized.

Model Layer

- User.js
- Workout.js
- Food.js

- Progress.js
- Recommendation.js

Controller Layer

- authController.js
- workoutController.js
- foodController.js
- progressController.js
- dashboardController.js
- recommendationController.js
- aiController.js

View / Route Layer

- authRoutes.js
- workoutRoutes.js
- foodRoutes.js
- progressRoutes.js
- dashboardRoutes.js
- recommendationRoutes.js
- aiRoutes.js

Supporting layers

- middleware/auth.js
- middleware/errorHandler.js
- services/aiRecommendation.js
- services/geminiService.js
- config/db.js

10. Project Setup and Configuration

Project Structure

- AI-FitTrack-API/
- server/
- server/client/
- server/config/
- server/controllers/
- server/middleware/
- server/models/
- server/routes/
- server/services/
- server/package.json
- server/package-lock.json
- server/server.js

Environment Security

The project contains a .env file locally for environment configuration. The repository uses .gitignore rules that exclude .env and node_modules from Git tracking. Secret values should remain local and must not be placed in the public repository.

Package installation and execution should follow the actual package.json scripts and environment configuration of the completed project. The supplied reference template contains example commands and example environment variables, but those examples are not copied as actual project values where the project report does not confirm them.

11. Backend Development

- Express server configuration.
- CORS configuration for frontend communication.
- MongoDB/Mongoose connection.
- Authentication and authorization checks.
- Progress, workout and food routes.
- Dashboard aggregation.
- Recommendation handling.
- Error responses.
- REST API communication using JSON request/response data.

AI Recommendation Service

The project contains AI-related service/controller files including aiController.js, aiRecommendation.js and geminiService.js. The supplied project documentation describes the AI capability as an AI-powered fitness recommendation module. The exact provider request/response schema and cost information are not specified in the source report, so those details are intentionally not invented here.

12. Database Development

MongoDB Atlas is used as the cloud database. The application stores structured JSON-like documents in collections corresponding to the main modules.

- users
- workouts
- progresses
- foods
- recommendations

Relationships

- One User → many Workouts
- One User → many Progress records
- Food → nutrition information
- Recommendation → recommendation information

13. Project Execution

The project is executed by running the backend and opening the browser-based client. The frontend communicates with the REST API and MongoDB-backed backend. Environment variables must be configured locally before execution.

Execution flow

- Open the AI-FitTrack-API project in Visual Studio Code.
- Open the server folder in the terminal.
- Install the dependencies using the project's package.json configuration.
- Ensure the local .env configuration is present and valid.
- Start the backend using the project's configured start command.
- Open the frontend and verify registration/login.
- Test progress, workout and food operations.
- Verify MongoDB Atlas collections and stored records.
- Verify dashboard statistics and AI recommendation output.

14. REST API Structure

Operation	HTTP Method	Purpose
Register	POST	Create user account
Login	POST	Authenticate user
Dashboard	GET	Retrieve dashboard information
Progress	GET / POST / PUT / DELETE	Manage progress records
Workouts	GET / POST / PUT / DELETE	Manage workout records
Workout Search	GET	Search workout records
Foods	GET / POST / PUT / DELETE	Manage food records

Protected operations use an Authorization header containing the JWT token.

15. Frontend Implementation

The frontend is organized around a browser-based application page containing registration, login, dashboard, progress, workout and food sections. HTML elements are connected to JavaScript functions using IDs and button click handlers. The stylesheet provides a card-based layout and responsive sizing.

- registerUser()
- loginUser()
- getDashboard()
- calculateBMI()
- addProgress() / getProgress()
- addWorkout() / getWorkouts() / searchWorkouts()
- addFood() / getFoods() / searchFoods()
- Edit and delete functions for stored records.

16. Authentication and Security

- Registration validation.
- Login validation.

- JWT token handling.
- Authorization header for protected API requests.
- Password protection through the backend authentication mechanism.
- Unauthorized access is rejected by the backend.
- Local .env and node_modules are excluded from Git tracking through .gitignore.

17. Progress Tracking and BMI

The Add Progress module accepts weight, height and date. BMI is calculated automatically from weight and height. The height is converted from centimetres to metres and the weight in kilograms is divided by the square of height in metres. The result is shown with two decimal places.

- Below 18.5 – Underweight
- 18.5 to below 25 – Normal
- 25 to below 30 – Overweight
- 30 and above – Obese

18. Workout Management

The workout module maintains exercise activity history. Each workout contains a name, category, duration, calories burned and workout date. Users can add, view, search, edit and delete workout records. Successful changes refresh the workout list and dashboard statistics.

19. Food and Nutrition Records

The food module stores food name, calories, protein, carbohydrates, fats and serving size. Users can add food records and view or search the food list. Edit and delete operations are also provided.

20. Dashboard and Recommendations

The dashboard summarizes the user's stored information. It includes total workouts, total calories burned, total foods, latest weight, latest height, latest BMI, BMI category and recommendation information. The AI Recommendation section displays recommendation text returned by the backend/AI functionality.

21. Validation and Error Handling

- Required-field validation.
- Numeric validation for weight, height, duration and calories.
- Authentication checks before protected operations.
- HTTP response status checking.
- Readable success/error messages.
- Console logging for debugging connection failures.

22. Testing Methodology

The backend APIs were tested using Thunder Client after starting the backend and connecting the application to MongoDB Atlas. Testing covers authentication, database operations,

progress, workouts, foods and dashboard-related requests.

- Check backend root/API response.
- Register a test user.
- Login and verify token response.
- Test protected dashboard request.
- Add and retrieve progress.
- Add, retrieve, search, update and delete workouts.
- Add and retrieve food records.
- Verify MongoDB Atlas collections and stored documents.
- Verify frontend displays returned data.

23. Testing Results

Test Area	Expected Result	Status
Registration	User account created	Completed
Login	Token returned and dashboard accessible	Completed
Progress	Record stored and BMI displayed	Completed
Workout CRUD	Create/read/update/delete works	Completed
Workout search	Matching records displayed	Completed
Food records	Food information displayed	Completed
MongoDB Atlas	Collections and documents visible	Completed
Dashboard	Counts and latest progress shown	Completed

Project Screenshot 1

Frontend – AI FitTrack Main Page

AI FitTrack

Personal Fitness Management System

Register

Register

Login

Login

Dashboard

New Dashboard

Total Workouts

0

Calories Burned

0

Total Foods

0

Latest Progress

Weight : N/A
Height : N/A
BMI: N/A
BMI Category: N/A
BMI Recommendation: No recommendation available.

AI Recommendation

No recommendation available.

 **Add Progress**

Project Screenshot 2

Registration and Login

Register

Register

Login

Login

Project Screenshot 3

Login Success and Dashboard

The screenshot displays the FitTrack application interface. At the top, there is a 'Login' section with two input fields: the first contains the email 'semu123@gmail.com' and the second contains masked characters '*****'. Below these fields is a dark 'Login' button. The bottom section is the 'Dashboard', which features a 'View Dashboard' button and three summary cards. The first card shows 'Total Workouts' as 6, the second shows 'Calories Burned' as 1150, and the third shows 'Total Foods' as 4.

Login

semu123@gmail.com

Login

Dashboard

View Dashboard

Total Workouts	Calories Burned	Total Foods
6	1150	4

Project Screenshot 4

Add Progress – Automatic BMI



The screenshot shows a web form titled "Add Progress" with a small icon to the left of the title. The form contains four input fields stacked vertically. The first field contains the number "64". The second field contains "170" and is highlighted with a thick black border. The third field contains "22.15". The fourth field contains a date placeholder "dd-mm-yyyy" and a small calendar icon on the right. Below the input fields is a dark button with the text "Add Progress".

Add Progress

64

170

22.15

dd-mm-yyyy

Add Progress

Project Screenshot 5

Progress History

Progress History

Show Progress

Progress

Weight: 64 kg

Height: 170 cm

BMI: 22.15

Date: 27/9/2026

Edit

Delete

Progress

Weight: 64 kg

Height: 170 cm

BMI: 22.15

Date: 27/9/2026

Edit

Delete

Progress

Weight: 64 kg

Height: 170 cm

BMI: 22.15

Date: 27/9/2026

Edit

Delete

Progress

Weight: 64 kg

Height: 170 cm

BMI: 22.15

Date: 27/9/2026

Edit

Delete

Progress

Weight: 64 kg

Height: 170 cm

BMI: 22.15

Date: 27/9/2026

Edit

Delete

Project Screenshot 6

Add Workout

 **Add Workout**

Project Screenshot 7

Workout List and Search

 MyWorkouts

Search

Show AllWork outs

Running

Category: Cardio

Duration: 30 minutes

Calories: 250

Date: 27/9/2026

Edit

Delete

Walking

Category: Cardio

Duration: 20 minutes

Calories: 100

Date: 27/9/2026

Edit

Delete

Project Screenshot 8

MongoDB Atlas – FitTrack Collections

The screenshot displays the MongoDB Atlas Data Explorer interface. The left sidebar shows the 'Data Explorer' section with 'My Queries' and 'Data Modeling' options. Under 'CLUSTERS', the 'fittrack' cluster is selected, showing a tree view of collections: 'foods', 'progresses', 'recommendations', 'users', and 'workouts'. The main panel shows a table of collections for the 'fittrack' cluster. The table has columns for 'Collection name', 'Properties', 'Storage size', 'Data size', 'Documents', 'Avg. document size', 'Indexes', and 'Total index size'. The data is as follows:

Collection name	Properties	Storage size	Data size	Documents	Avg. document size	Indexes	Total index size
foods	-	36.86 KB	683.00 B	4	170.00 B	1	36.86 KB
progresses	-	36.86 KB	110 KB	6	138.00 B	1	36.86 KB
recommendations	-	36.86 KB	104 KB	3	348.00 B	1	36.86 KB
users	-	36.86 KB	756.00 B	4	189.00 B	2	73.73 KB
workouts	-	36.86 KB	132 KB	7	188.00 B	1	36.86 KB

At the bottom of the interface, there is a status bar showing 'Alert System Status: All Good' and '©2024 MongoDB, Inc. Status Terms Privacy Atlas Blog Contact Sales'. The Windows taskbar at the very bottom shows the date as 27-09-2024 and the time as 15:46.

Project Screenshot 9

Add Food and Food List

 **Add Food**

 **Food List**

 Apple

Calories: 52
Protein: 0.3 g
Carbohydrates: 14 g
Fats: 0.2 g
Serving Size: 100g

 banana

Calories: 52
Protein: 0.2 g
Carbohydrates: 56 g

Project Screenshot 10

AI Recommendation

Latest Progress

Weight: 64 kg
Height: 170 cm
BMI: 22.15
BMI Category: Normal
BMI Recommendation: Your BMI is in the normal range. Continue maintaining balanced nutrition and regular physical activity.

AI Recommendation

Your BMI is in the normal range. Maintain a balanced diet and regular exercise. Try to maintain a more consistent work out schedule during the week. Gradually increase your physical activity according to your fitness level.

Project Screenshot 11

Final Dashboard – Results

Login

Login

Dashboard

View Dashboard

Total Workouts
6

Calories Burned
1150

Total Foods
4

Latest Progress

Weight: 64 kg
Height: 170 cm
BMI: 22.15
BMI Category: Normal
BMI Recommendation: Your BMI is in the normal range. Continue maintaining balanced nutrition and regular physical activity.

AI Recommendation

Your BMI is in the normal range. Maintain a balanced diet and regular exercise. Try to maintain a more consistent workout schedule during the week. Gradually increase your physical activity according to your fitness level.

 Add Progress

Add Progress

Project Screenshot 12

Final Application – Complete View

AI FitTrack

Personal Fitness Management System

Register

Register

Login

Login

Dashboard

View Dashboard

Your Workouts

5

Calories Burned

950

Total Goals

4

Latest Progress

Weight: 85kg

Height: 1.80m

BMI: 26.75

BMI Category: Normal

BMI Recommendation: Your BMI is in the normal range. Continue maintaining a balanced diet and regular physical activity.

AI Recommendation

You often have a rest day. AI often suggests a rest day to allow your body to recover. Try to maintain a more consistent workout schedule to optimize your health and fitness goals.

Add Progress

1%

11/11/2023

Add Progress

Progress History

Show Progress

Click "Show Progress" to view your progress.

Add Workout

11/11/2023

Add Workout

My Workouts

Search

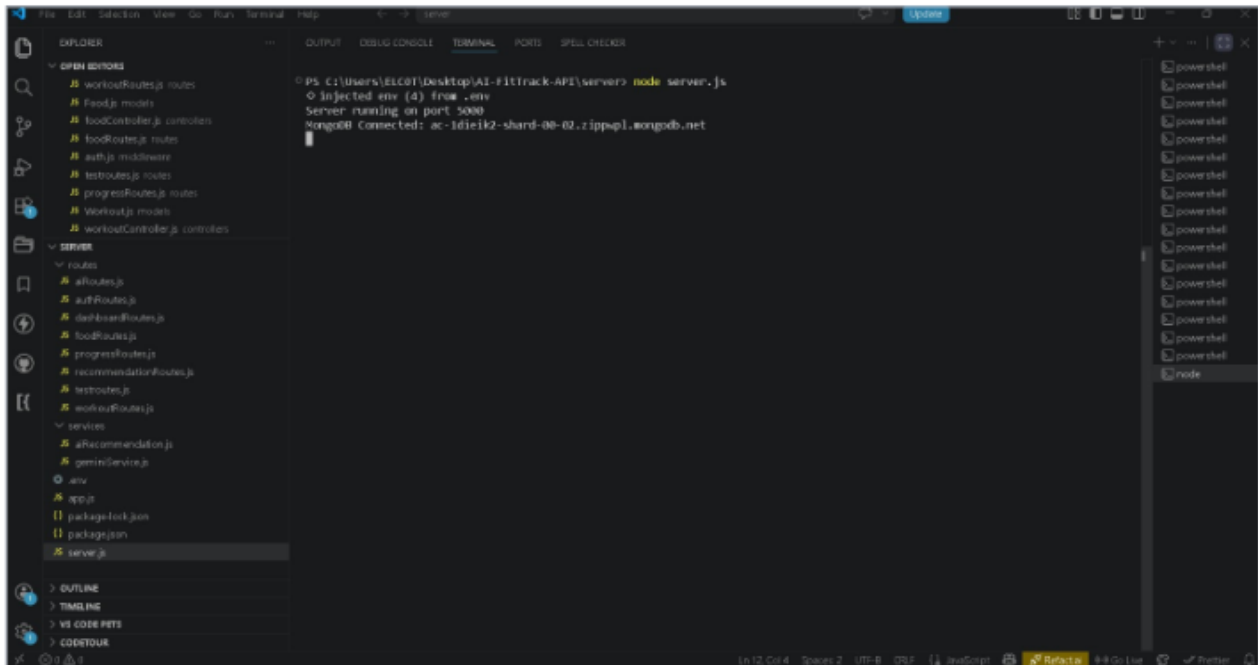
Show My Workouts

Click "Show My Workouts" to view your workouts.

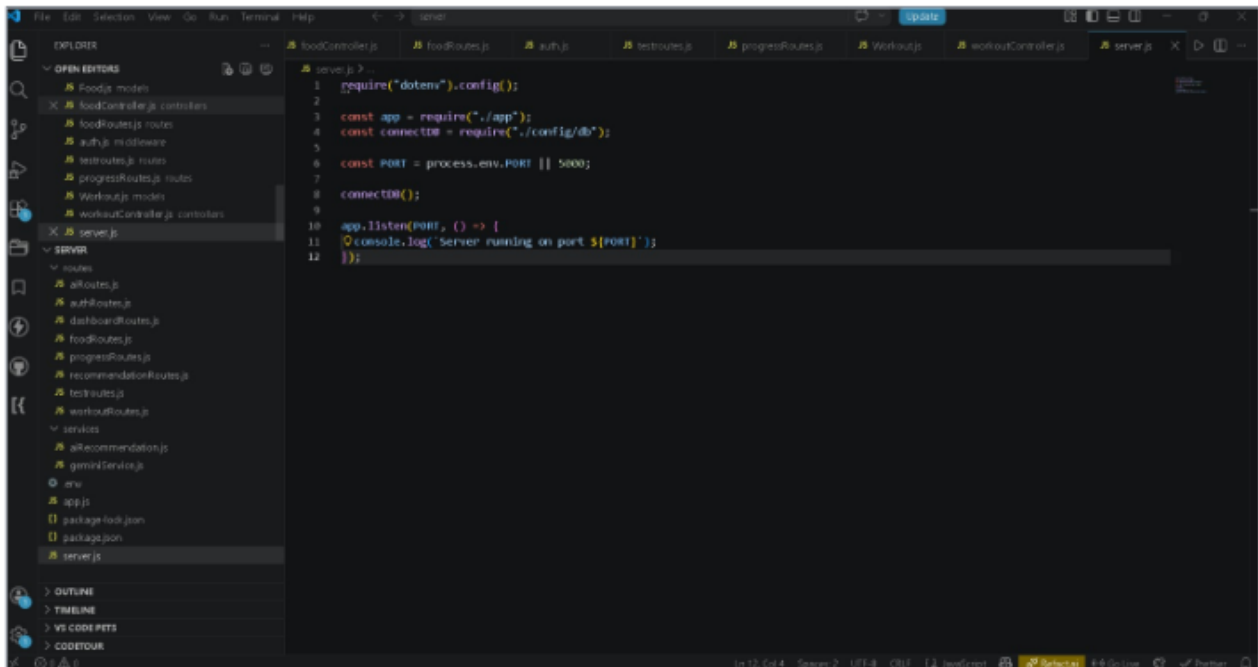
Supplementary Screenshots

Backend – Server and Database Connection

A. Backend server running on port 5000 with MongoDB Atlas connected



B. server.js – Express server entry point and database connection call



Project Screenshot 12

Final Application – Complete View

Project Screensnot 1

Frontend – AI FitTrack Main Page

AI FitTrack

Personal Fitness Management System

Register

Enter your name

Enter your email

Enter your password

Register

Login

Enter your email

Enter your password

Login

Dashboard

View Dashboard

Total Workouts

0

Calories Burned

0

Total Foods

0

Latest Progress

Weight : N/A

Height : N/A

BMI: N/A

BMI Category: N/A

22. Conclusion

AI FitTrack successfully combines a browser-based frontend, REST backend and MongoDB Atlas database into a single fitness management system. The application provides authentication, progress tracking, automatic BMI calculation, workout management, food records and dashboard summaries.

The project demonstrates practical use of HTML, CSS, JavaScript, Node.js, Express.js, MongoDB, Mongoose, JWT-based authentication and API testing with Thunder Client. The modular design also provides a foundation for adding more fitness-related functionality in the future.

23. Future Enhancements

The current system can be extended without changing its basic architecture.

- Charts for weight and workout progress.
- More detailed nutrition analytics.
- Workout reminders and notifications.
- Wearable-device integration.
- Role-based administration.
- Improved AI-generated plans and progress insights.
- Exporting progress reports.
- Cloud deployment of frontend and backend.
- Additional validation and automated testing.

24. Project Documentation Summary

This report has been prepared as a project documentation document for AI FitTrack. It describes the actual application modules, technology stack, database structure, frontend behaviour, backend communication and testing workflow.

Screenshots of the working application, the MongoDB Atlas database and the backend server are included in the screenshot section of this document as evidence of the completed project.

End of Project Documentation